Introduction to Causal Inference for Software Engineering

A 90-minute hands-on tutorial

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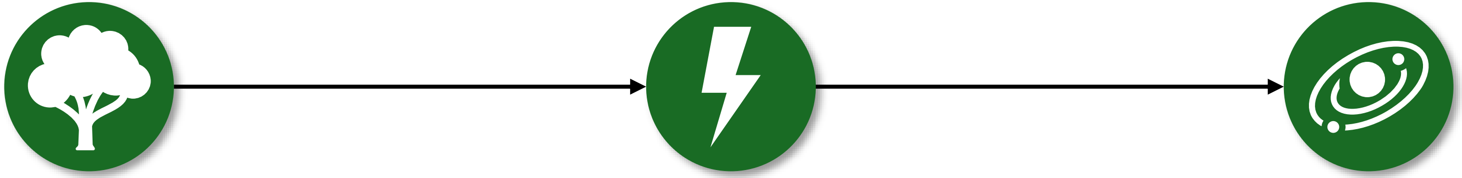
Section 2d

Causal Models as Requirements Engineering Activity

What is the problem?

- RE struggles to translate high-level requirements into data needs.
- One reason for this under specification is that operational context is often **implicitly assumed in the data collection.**



Operational Context
implicitly assumed in data collection

Brittle Models
that fail to generalize

Low Transferability
into unknown contexts

Ahmad, K., Abdelrazek, M., Arora, C., Bano, M., & Grundy, J. (2023). Requirements engineering for artificial intelligence systems: A systematic mapping study. *Information and Software Technology*, *158*, 107176.
Mitchell, S., Potash, E., Barocas, S., D'Amour, A., & Lum, K. (2021). Algorithmic fairness: Choices, assumptions, and definitions. *Annual review of statistics and its application*, *8*(1), 141-163.
D'Amour, A., Heller, K., Moldovan, D., Adlam, B., Alipanahi, B., Beutel, A., ... & Sculley, D. (2022). Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, *23*(226), 1-61.

Causal models are a formal representation of prior knowledge

Using the principles of Statistical Causal Inference in RE allows to ...

- … derive data and model requirements from prior knowledge about the operational context.
- … enable documentation of these prior assumptions.
- … allow for a systematic determination of testable conditions.
- … **use of causal modelling as a Requirement Engineering activity.**

Use case: Arcs in DC switching gear

- ML-based classification (outcome) of arc faults (exposure) in DC switching gear.
- System uses a current sensor for detecting the occurrence of an arc.

Functional Requirement:

GIVEN *a low voltage DC system in normal operation*
WHEN *an arc occurs*
THEN *an alarm should be triggered.*

Methodolgy

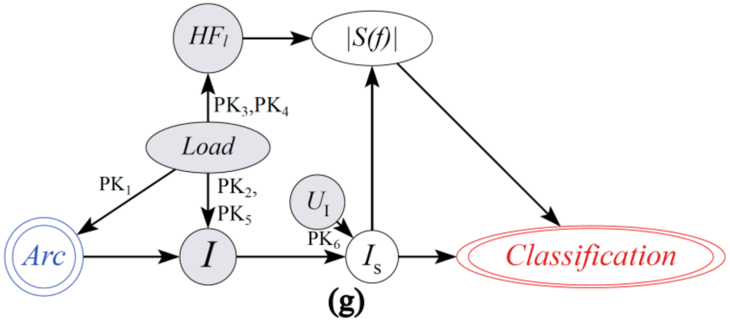
- Three exploratory workshops with two engineers and two researchers.
- We introduced to the concept of causal modelling and d-separation.
- Prior knowledge was formalised by identifying causal mechanisms and updating the causal model.

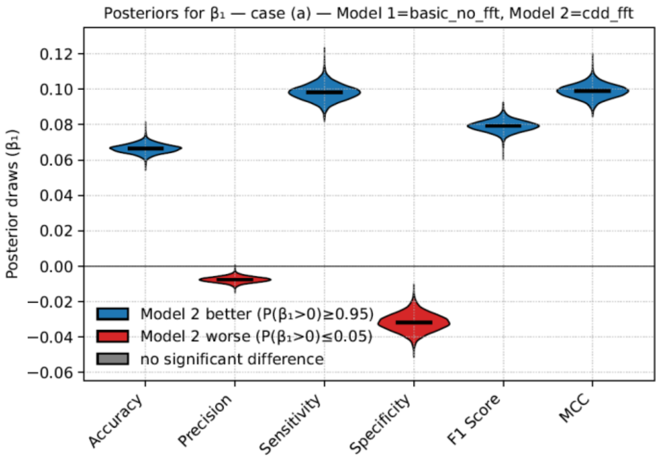
Interesting observation: The structured approach to formalising prior knowledge as causal graphs allowed them to ”discover” more prior knowledge.

- We used then d-separation to close non-causal paths between exposure (i.e., the occurrence of a fault), and the outcome (i.e., the classification result).
- This resulted in data and model requirements, as well as testable conditions.

Did causal modelling help?

- We formalised six elements of prior knowledge into three additional data requirements and one model requirement.
- We compared an existing model with a model following causality-driven data requirements using a dataset that simulates a context change.
- **The false negative rate decreases significantly (increase in sensitivity threefold to specificity).**
- Overall, the causality driven model generalises better to an unknown context.





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Open research questions on causal modelling as RE activity

- How can causal models complement natural language requirements? How must causal modelling be adopted to RE?
- When does a causal graph include ”enough” prior knowledge? What is a sufficient set of variables in a causal graph for a given use case?
- How can the modularisation of cause-effect mechanisms help in modularisation of ML software systems?
- How can causal models serve as ”boundary object” between different stakeholders?
- To what degree can data requirements from causal models enhance ML robustness and reduce data needs?
- How can independence conditions be translated into testing strategies?